

TLS_Pie troubleshooting guide

Printable fault-finding reference

What changed from the original

- Arduino record outputs moved from A4/A3 to D7/D8 to avoid display conflicts.
- Pi scripts now wait for Arduino trigger pulses on GPIO17 and GPIO27.
- The capture directory is now created before tcpdump starts.

Quick wiring summary

- Arduino D7 -> Pi GPIO17 (record start)
- Arduino D8 -> Pi GPIO27 (record stop)
- Arduino GND -> Pi GND
- A0/A1/A2/D2 -> buttons -> GND
- D3/D5/D6 -> stepper driver inputs
- Use a separate motor supply for the stepper driver motor power.

Where to test

- 5V rail and GND on Arduino and Pi
- Button input pins A0, A1, A2, D2
- Trigger pins D7 and D8
- Stepper driver pins D3, D5, D6
- Pi GPIO17 and GPIO27

Expected behavior

- Display should start and show the menu.
- Button press should change the selection.
- Starting a scan should briefly pull D7 LOW.
- Stopping a scan should briefly pull D8 LOW.
- A .pcap file should be created by tcpdump.

Quick checklist

- Power present on Arduino and Pi
- Common ground connected
- Buttons pull the pins low when pressed
- Stepper driver receives step pulses
- Pi receives the Arduino trigger pulses
- Capture folder exists and tcpdump runs

If anything fails, start at the top and verify power, ground, then buttons, then trigger pulses, then the stepper driver and Pi capture.